SUVAS T V

+91 8248043716 tvsuvas@gmail.com LinkedIn GitHub Pollachi, Tamil Nadu

PROFESSIONAL SUMMARY

Highly motivated ECE undergraduate (CGPA: 9.69) with hands-on experience in FPGA design, embedded systems, and Edge AI. Eager to contribute to real-world engineering projects through an internship in VLSI, FPGA, or embedded systems. Committed to bridging academic knowledge with practical industry exposure to develop impactful solutions.

EDUCATION

SRM Institute of Science and Technology, Tiruchirappalli 2024 – 2028 (Expected)
Bachelor of Technology in Electronics and Communication Engineering CGPA: 9.69/10.00

Honey Bunch Senior Secondary School (CBSE), Pollachi
Class XII — 88% | *Class X — 96.2%*

TECHNICAL SKILLS

Programming Languages: C, Python
Hardware Description Languages: Verilog HDL, VHDL, SystemVerilog
Hardware Design: FPGA Design, VLSI Design, Embedded Systems, Digital Logic Design, Edge AI, Signal Processing
EDA & Development Tools: Xilinx Vivado, ModelSim, MATLAB, LTspice, Git
Hardware / Platforms: FPGA (Xilinx/AMD), Arduino, Raspberry Pi

RELEVANT COURSEWORK

Digital Electronics • VLSI Design • Embedded Systems • Signals & Systems • FPGA Design

PROJECTS

Real-Time Cardiac Fatigue Detection using HRV on FPGA (Edge AI) *Verilog HDL, FPGA (Xilinx), Vivado* GitHub

- Developed an FPGA-based Edge AI system for real-time cardiac fatigue detection using Heart Rate Variability (HRV) signals.
- Designed RTL hardware modules for signal preprocessing, HRV feature extraction, and on-device MLP inference.

Automated Street Lighting System using LDR and Arduino *Arduino, LDR Sensor, Embedded C* GitHub

- Designed an automatic lighting system that adapts to ambient light using LDR sensing and Arduino control logic; filed as an Indian patent application.
- Developed real-time switching firmware to reduce energy consumption in smart city environments.

Spiking Neural Network using LIF Neuron Model Simulation *Python, NumPy, MATLAB* GitHub

- Simulated a Leaky Integrate-and-Fire (LIF) spiking neural network to study neuromorphic computing principles.
- Analyzed temporal spike patterns and membrane dynamics for event-driven computation research.

ACHIEVEMENTS

Secured 2nd Prize in BUILD N BLIND, NIT Tiruchirappalli — a reverse-engineering challenge involving incomplete circuits and abstract functional descriptions, requiring creative circuit realization under communication constraints.

PATENT

Published Indian Patent Application — Automated Street Lighting System using Light Dependent Resistor (LDR) and Arduino, Application No. 202641024982 A

LANGUAGES

English • Tamil • Telugu • Hindi